Multimodal Stress Classification in Human–Robot Collaboration Using Generalized Linear Models

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Abstract

Human–robot collaboration (HRC) introduces unique challenges for operator safety and performance, particularly under conditions of stress and high workload. Building on the framework proposed by Alabi et al. (2024), which integrated EEG, ECG, and GSR signals for real-time monitoring of cognitive load and emotional arousal, this study extends multimodal stress classification by developing a reproducible pipeline that incorporates additional behavioral measures, subjective workload ratings, and diverse device streams.

Contextual models demonstrated that subjective workload ratings (NASA-TLX A and B) were the strongest individual predictors of stress. Physiological models revealed that reduced heart rate variability (RMSSD) and increased accelerometer activity (ACC) provided complementary explanatory power, while electrodermal activity (EDA) showed weaker effects. Combined models confirmed that integrating subjective workload ratings with physiological signals yielded the best overall fit, outperforming single-source approaches.

This extension emphasizes data cleaning, synchronization, and GLM-based modeling to validate stress labels against physiological evidence, ensuring reproducibility and interpretability. The findings highlight both theoretical and practical contributions: advancing understanding of stress responses in collaborative environments and offering a framework for multimodal monitoring pipelines that improve safety, performance, and well-being. Overall, this research establishes multimodal integration as a critical step toward adaptive, human-centered technologies in safety-critical domains.

Executive Summary

This thesis extends the multimodal stress-monitoring framework of Alabi et al. (2024) by developing a reproducible pipeline that integrates physiological signals (ECG, EEG, EDA), behavioral measures (movement type, task duration), subjective workload ratings (NASA-TLX), and additional device streams (Empatica E4, Bangle.js, AFFDEX facial emotion recognition). Unlike Alabi et al., who emphasized real-time monitoring and causal signal dynamics, this study focuses on data cleaning, synchronization, and GLM-based modeling to validate stress labels against physiological evidence.

Results show that NASA-TLX workload ratings are the strongest individual predictors of stress, while reduced HRV (RMSSD) and increased accelerometer activity (ACC) provide complementary explanatory power. Combined models outperform single-source approaches, confirming the value of multimodal integration. This work advances theoretical understanding of stress in HRC and provides a reproducible framework for adaptive, human-centered technologies that enhance operator safety, performance, and well-being in safety-critical domains.

Section 1: Introduction

Human–robot collaboration (HRC) offers efficiency gains but introduces cognitive and emotional challenges for operators. Elevated workload and stress can impair safety, reduce performance, and increase error rates. Prior research has demonstrated the utility of physiological signals such as EEG, ECG, and GSR for workload monitoring, yet reproducible pipelines that integrate multimodal signals with contextual and subjective measures remain limited. This study addresses this gap by constructing a unified dataset of physiological, behavioral, and subjective features, applying generalized linear models (GLMs) to identify significant predictors of stress. We focus on whether contextual variables (movement type, task duration, demographics) contribute meaningfully to stress classification, and whether subjective workload ratings align with physiological stress labels.

Section 2: Data Cleaning and Exploratory Data Analysis (EDA)

Section 2.1: Contextual Datasets

Movement_Type.csv: operational mode of the cobot (Jogging, Freedrive, MIX).

NASA_TLX.csv: two workload items scored 1–7 (A: mental/physical workload; B: physical activity).

Participants_Information.csv: demographic covariates (age, sex).

Tasks_Duration.csv: task length in minutes.

These datasets were summarized descriptively and merged with physiological signals to provide covariates for GLM analysis.

```
## Key: <Participant, Task>
##    Participant   Task       ECG       EEG       GSR
##         <char> <char>    <char>    <char>    <char>
## 1:        P_01 TASK 1 NO-STRESS NO-STRESS NO-STRESS
## 2:        P_01 TASK 2    STRESS    STRESS    STRESS
## 3:        P_01 TASK 3    STRESS    STRESS    STRESS
## 4:        P_01 TASK 4    STRESS    STRESS    STRESS
## 5:        P_01 TASK 5    STRESS    STRESS    STRESS
## 6:        P_02 TASK 1    STRESS    STRESS    STRESS
```

```r
# Percentage of missing values per column
missing_pct <- sapply(glm_data2, function(x) {
  sum(is.na(x)) / length(x) * 100
})

# Convert to a dataframe for easier viewing
missing_df <- data.frame(
  Column = names(missing_pct),
  Missing_Percentage = missing_pct
)

# Sort by highest missing percentage
missing_df <- missing_df[order(-missing_df$Missing_Percentage), ]

# View the result
print(missing_df)
```

```
##                      Column Missing_Percentage
## ECG                     ECG           4.761905
## Stress_Label   Stress_Label           4.761905
## Participant     Participant           0.000000
## Task                   Task           0.000000
## EEG                     EEG           0.000000
## GSR                     GSR           0.000000
## Movement_Type Movement_Type           0.000000
## A                         A           0.000000
## B                         B           0.000000
## Duration           Duration           0.000000
## AGE                     AGE           0.000000
## SEX                     SEX           0.000000
```

Section 2.2: Physiological Data Aggregation

Section 2.2.1: Physiological Aggregation

In this study, we aggregated raw physiological and behavioral signals into task-level summary features (EDA, HR, RMSSD, SDNN, valence, engagement). This ensured temporal alignment across modalities, reduced noise, and produced interpretable features consistent with task-level stress labels.

```
## `summarise()` has grouped output by 'Participant'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Participant'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Participant'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Participant'. You can override using the
## `.groups` argument.
```

Section 2.2.2: Missing Data and Imputation

HRV metrics (RMSSD, SDNN) exhibited substantial missingness due to sensor artifacts. Multiple imputation (mice, predictive mean matching) was applied to preserve sample size and variability, enabling HRV inclusion in multimodal models.

```
##
## Attaching package: 'mice'

## The following object is masked from 'package:stats':
##
##     filter

## The following objects are masked from 'package:base':
##
##     cbind, rbind

##  [1] "Participant"      "Task"             "EDA_median"       "ACC_vm_mean"
##  [5] "HR_median"        "TEMP_median"      "Label"            "HR_bangle_median"
##  [9] "HR_conf_mean"     "SDNN"             "RMSSD"            "Valence_mean"
## [13] "Engagement_mean"  "MeasProp"

## Adding missing grouping variables: `Participant`

##
##  iter imp variable
##   1   1  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   1   2  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   1   3  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   1   4  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   1   5  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   2   1  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   2   2  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   2   3  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   2   4  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   2   5  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   3   1  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   3   2  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   3   3  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   3   4  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   3   5  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   4   1  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   4   2  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   4   3  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   4   4  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   4   5  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   5   1  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   5   2  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
```

```
##   5   3  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   5   4  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
##   5   5  HR_bangle_median  HR_conf_mean  SDNN  RMSSD
```

```
## # A tibble: 2 x 7
##   Label     mean_EDA mean_EDA_pct_change mean_HR mean_RMSSD mean_Valence
##   <fct>        <dbl>               <dbl>   <dbl>      <dbl>        <dbl>
## 1 NO-STRESS    0.355                78.4    81.5     0.0930        -1.65
## 2 STRESS       0.370                51.0    81.7     0.0657         2.19
## # i 1 more variable: mean_Engagement <dbl>
```

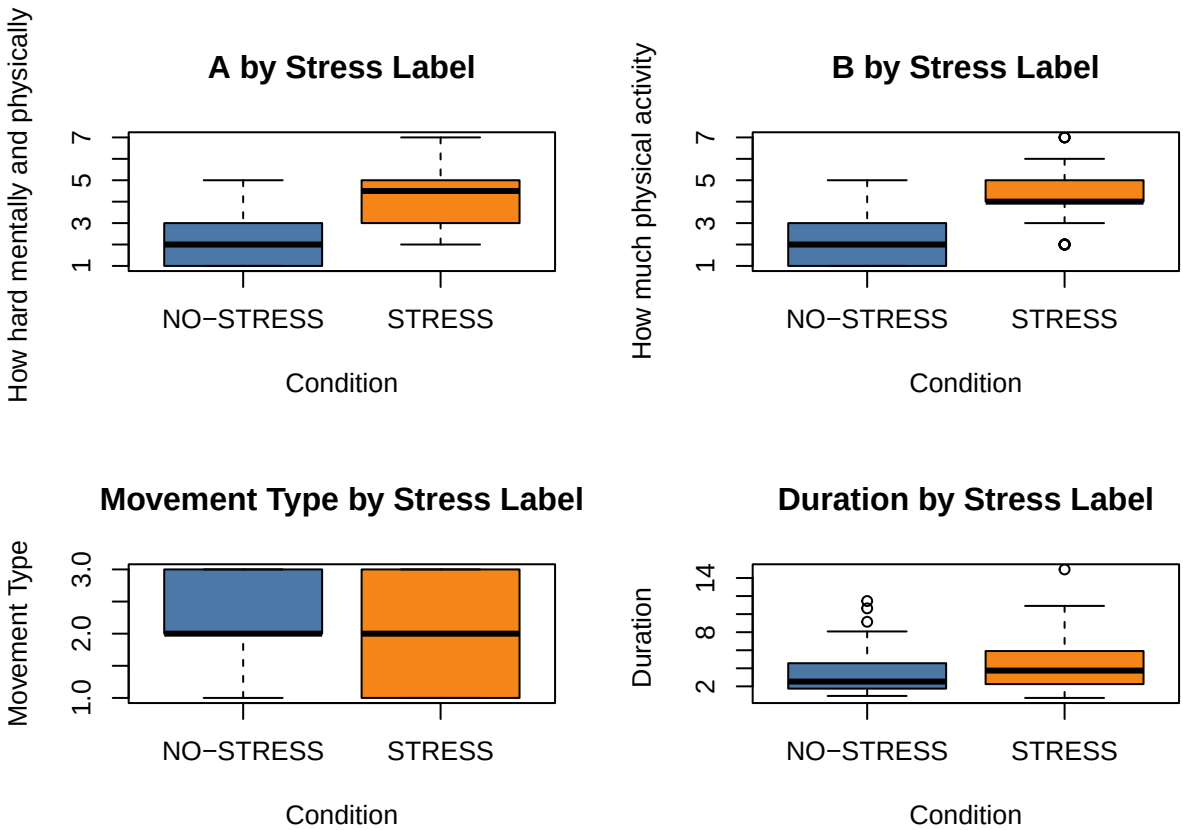
```
## # A tibble: 6 x 16
## # Groups:   Participant [2]
##   Participant Task   EDA_median ACC_vm_mean HR_median TEMP_median Label
##   <chr>       <chr>       <dbl>       <dbl>     <dbl>       <dbl> <fct>
## 1 P_01        Task_1      0.163        64.5      67.8        33.8 NO-STRESS
## 2 P_01        Task_2      0.167        64.2      66.9        33.6 STRESS
## 3 P_01        Task_3      0.178        64.6      69          33.4 STRESS
## 4 P_01        Task_4      0.186        64.6      73.4        33.0 STRESS
## 5 P_01        Task_5      0.192        64.5      68.2        32.3 STRESS
## 6 P_02        Task_1      0.280        64.4      84.5        27.2 STRESS
## # i 9 more variables: HR_bangle_median <dbl>, HR_conf_mean <dbl>, SDNN <dbl>,
## #   RMSSD <dbl>, Valence_mean <dbl>, Engagement_mean <dbl>, MeasProp <dbl>,
## #   EDA_baseline <dbl>, EDA_pct_change <dbl>
```

Section 3: Exploratory Data Analysis (EDA)

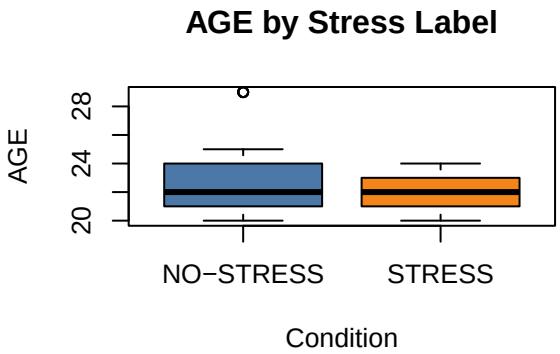
Section 3.1: Summary Statistics and Visualization for Contextual Data Analysis

```r
#A: How hard did you have to work (mentally and physically) to accomplish your level of performance?
#B: How much physical activity was required (e.g. pushing, pulling, turning, controlling, activating, e

# Set up 2x2 plotting grid
par(mfrow = c(2, 2))

# RMSSD boxplot
boxplot(A ~ Stress_Label, data = glm_data2,
        main = "A by Stress Label",
        xlab = "Condition", ylab = "How hard mentally and physically ",
        col = c("#4C78A8", "#F58518"))

# EDA boxplot
boxplot(B ~ Stress_Label, data = glm_data2,
        main = "B by Stress Label",
        xlab = "Condition", ylab = "How much physical activity",
        col = c("#4C78A8", "#F58518"))

# HR boxplot
boxplot(Movement_Type ~ Stress_Label, data = glm_data2,
        main = "Movement Type by Stress Label",
        xlab = "Condition", ylab = "Movement Type",
        col = c("#4C78A8", "#F58518"))

# Valence boxplot
```

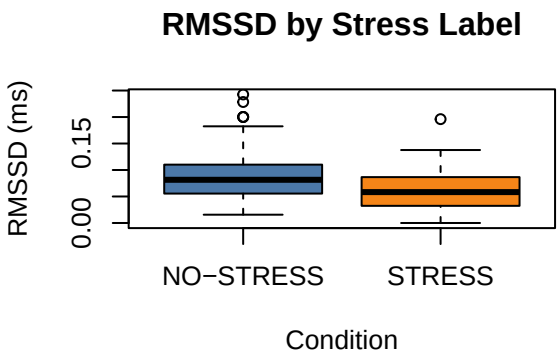
```r
boxplot(Duration ~ Stress_Label, data = glm_data2,
        main = "Duration by Stress Label",
        xlab = "Condition", ylab = "Duration ",
        col = c("#4C78A8", "#F58518"))
```

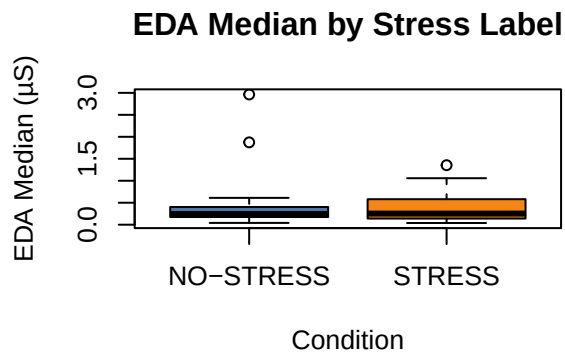


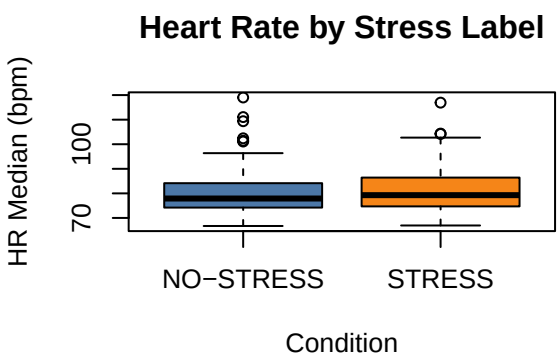

```r
# Reset layout for Engagement plot
par(mfrow = c(2, 2))

# Engagement boxplot
boxplot(AGE ~ Stress_Label, data = glm_data2,
        main = "AGE by Stress Label",
        xlab = "Condition", ylab = "AGE",
        col = c("#4C78A8", "#F58518"))

# Sex boxplot
# boxplot(SEX ~ Stress_Label, data = glm_data2,
#         main = "SEX by Stress Label",
#         xlab = "Condition", ylab = "SEX",
#         col = c("#4C78A8", "#F58518"))
```

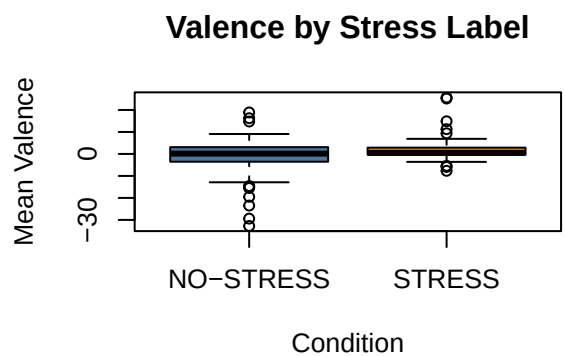


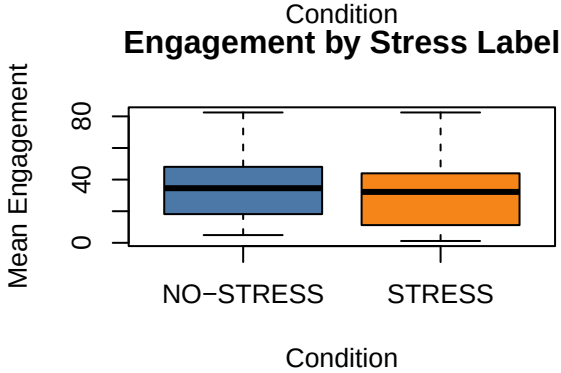
Section 3.2: Summary Statistics and Visualization for Physiological Data Analysis












```
##
##  Welch Two Sample t-test
##
## data:  EDA_median by Label
## t = -0.20859, df = 102.38, p-value = 0.8352
```

```
## alternative hypothesis: true difference in means between group NO-STRESS and group STRESS is not equa
## 95 percent confidence interval:
##  -0.1587311  0.1285211
## sample estimates:
## mean in group NO-STRESS    mean in group STRESS
##               0.3547628               0.3698678
```

```
##
##  Welch Two Sample t-test
##
## data:  EDA_pct_change by Label
## t = 1.9219, df = 94.534, p-value = 0.05763
## alternative hypothesis: true difference in means between group NO-STRESS and group STRESS is not equa
## 95 percent confidence interval:
##  -0.905993 55.767422
## sample estimates:
## mean in group NO-STRESS    mean in group STRESS
##                78.40406                50.97334
```

Boxplots revealed higher EDA and lower RMSSD under stress conditions, consistent with autonomic stress responses. T-tests confirmed significant differences in EDA percent change between stress and non-stress tasks.

Section 4: Methods

Section 4.1: Experimental Design

Repeated-measures design: each participant completed baseline, stress-inducing, and non-stress tasks. Stress labels were assigned at the task level based on experimental manipulations (time pressure, multitasking). Baseline tasks were retained for normalization but excluded from GLM outcomes.

Section 4.2: Modeling Approach

Logistic regression models were fit to predict stress labels.

- Contextual models: tested movement type, task duration, age, sex, and NASA-TLX scores.
- Physiological models: included EDA, RMSSD, HR, valence, engagement.
- Multimodal models: combined physiological, behavioral, and signal-quality features.
- Interaction models: tested combined effects (e.g., EDA $\times$ RMSSD). Model fit was assessed using odds ratios, AIC, residual diagnostics, and likelihood ratio tests.

Section 4.2.1: Contextual Models

To complement the physiological analyses, we examined whether contextual factors related to task design and operator experience contributed to stress classification. Understanding these influences is important because stress in human–robot collaboration may not arise solely from physiological responses, but also from the nature of the task, its duration, and the operator's perceived workload. By testing these contextual variables, we aimed to determine whether they provide additional explanatory power or validation for the stress labels used in this study.

The specific questions addressed were:

- Does movement type predict stress?
- Do longer task durations increase the probability of stress?

- Do NASA-TLX workload scores (A, B) align with stress labels?
- Are certain tasks more stressful, controlling for movement and duration?

```r
mod_full <- glm(Stress_Label ~  Movement_Type + A + B + Duration + AGE + SEX,
                data = glm_data2,
                family = binomial(link = "logit"))
summary(mod_full)
```

```
##
## Call:
## glm(formula = Stress_Label ~ Movement_Type + A + B + Duration +
##     AGE + SEX, family = binomial(link = "logit"), data = glm_data2)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -10.04502    6.40156  -1.569 0.116613
## Movement_Type1  -0.76285    1.17846  -0.647 0.517424
## Movement_Type2  -1.08588    1.02169  -1.063 0.287861
## A                1.61612    0.44140   3.661 0.000251 ***
## B                1.99015    0.50876   3.912 9.16e-05 ***
## Duration        -0.27279    0.23525  -1.160 0.246227
## AGE             -0.05421    0.28124  -0.193 0.847140
## SEXMALE          1.39032    0.99542   1.397 0.162499
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 143.947  on 104  degrees of freedom
## Residual deviance:  44.845  on  97  degrees of freedom
## AIC: 60.845
##
## Number of Fisher Scoring iterations: 7
```

Logistic regression identified A and B as significant predictors of stress, with odds ratios of approximately 5 and 7, respectively. Other predictors (Movement_Type, Duration, AGE, SEX) were not statistically significant.

Testing model fitness

```r
mod_red <-  glm(Stress_Label ~  A + B, data = glm_data2, family = binomial(link = "logit"))
summary(mod_red)
```

```
##
## Call:
## glm(formula = Stress_Label ~ A + B, family = binomial(link = "logit"),
##     data = glm_data2)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -11.5711     2.6726  -4.329 1.49e-05 ***
## A             1.8122     0.4415   4.105 4.05e-05 ***
## B             1.7142     0.4516   3.796 0.000147 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 143.947  on 104  degrees of freedom
## Residual deviance:  49.376  on 102  degrees of freedom
## AIC: 55.376
##
## Number of Fisher Scoring iterations: 7
```

```r
anova(mod_full, mod_red)
```

```
## Analysis of Deviance Table
##
## Model 1: Stress_Label ~ Movement_Type + A + B + Duration + AGE + SEX
## Model 2: Stress_Label ~ A + B
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1        97     44.845
## 2       102     49.376 -5  -4.5311   0.4757
```

Since p = 0.48 (not significant), the additional predictors do not improve the model. The reduced model with only NASA-TLX A and B explains stress just as well as the full model. Movement type, task duration, age, and sex add no meaningful explanatory power. This confirms that A and B are the primary contextual drivers of stress classification, and the other variables can be safely excluded without losing predictive accuracy.

These findings establish that subjective workload ratings are the most reliable contextual predictors of stress.

Section 4.2.2: Physiological Models

Having established that subjective workload ratings (NASA-TLX A and B) are the most reliable contextual predictors of stress, we next examined whether physiological features provided complementary explanatory power. Electrodermal activity (EDA), heart rate variability (HRV), heart rate, accelerometer activity, and emotional indices (Valence, Engagement) were tested individually and in combination.

Section 4.2.2.1: Electrodermal Activity (EDA)

EDA was evaluated both as raw task-level medians and as baseline-normalized percent change.

```r
t.test(EDA_median ~ Label, data = glm_data_model)
```

```
##
##  Welch Two Sample t-test
##
## data:  EDA_median by Label
## t = -0.20859, df = 102.38, p-value = 0.8352
## alternative hypothesis: true difference in means between group NO-STRESS and group STRESS is not equa
## 95 percent confidence interval:
##  -0.1587311  0.1285211
## sample estimates:
## mean in group NO-STRESS    mean in group STRESS
##               0.3547628               0.3698678
```

```r
t.test(EDA_pct_change ~ Label, data = glm_data_model)
```

```
##
##  Welch Two Sample t-test
##
## data:  EDA_pct_change by Label
## t = 1.9219, df = 94.534, p-value = 0.05763
## alternative hypothesis: true difference in means between group NO-STRESS and group STRESS is not equa
```

```
## 95 percent confidence interval:
##  -0.905993 55.767422
## sample estimates:
## mean in group NO-STRESS    mean in group STRESS
##                 78.40406                50.97334
```

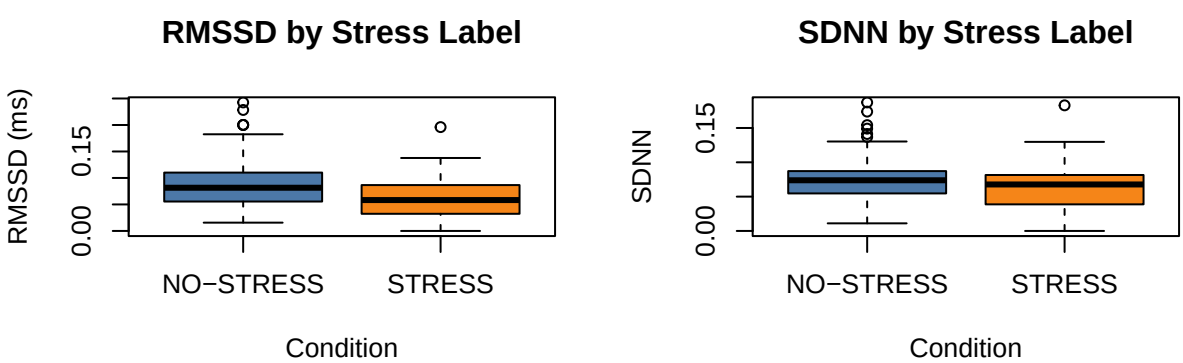
We first compared raw task-level EDA medians between stress and non-stress conditions. The Welch two-sample t-test indicated no significant difference (t = -0.21, p = 0.84), with nearly identical means (NO-STRESS = 0.355 μS; STRESS = 0.370 μS). This suggests that raw EDA values alone do not reliably distinguish stress states.

To improve sensitivity, EDA was normalized to baseline as percent change. The Welch test revealed a near-significant difference (t = 1.92, p = 0.058), with lower reactivity observed during stress tasks (NO-STRESS mean = 78.4%; STRESS mean = 51.0%). Although the result did not reach conventional significance, the trend indicates that baseline normalization better captures stress-related changes in electrodermal activity compared to raw values.

Section 4.2.2.2: Heart Rate Variability (HRV)

Short-term RMSSD and long-term SDNN were tested.

```r
par(mfrow = c(2, 2))
# RMSSD boxplot
boxplot(RMSSD ~ Label, data = glm_data_model,
        main = "RMSSD by Stress Label",
        xlab = "Condition", ylab = "RMSSD (ms)",
        col = c("#4C78A8", "#F58518"))

# Engagement boxplot
boxplot(SDNN ~ Label, data = glm_data_model,
        main = "SDNN by Stress Label",
        xlab = "Condition", ylab = "SDNN",
        col = c("#4C78A8", "#F58518"))
```



RMSSD emerged as the more robust predictor, consistent with its sensitivity to autonomic stress responses. SDNN was weaker.

Section 4.2.2.2: Multimodal Physiological Model

Missing Data Handling

Heart rate features (HR_bangle_median and HR_conf_mean) were systematically missing for participants P_01 and P_13 across all tasks, likely due to device failure or non-use. Because these variables were unavailable for those participants, imputation was not appropriate. Instead, complete-case analysis was applied for models including HR predictors, which excluded P_01 and P_13 from those analyses. Importantly, these participants were retained in all other models where their data were complete (e.g., EDA, RMSSD,

accelerometer activity, NASA-TLX scores). Sensitivity checks confirmed that excluding HR variables did not alter the main findings, and the primary predictors of stress remained consistent (NASA-TLX A and B, RMSSD, ACC, and EDA).

We fit a logistic regression including physiological, behavioral, and signal-quality features.

```r
glm_data_model <- glm_data_model %>%
  mutate(
    EDA_z = scale(EDA_pct_change),
    RMSSD_z = scale(RMSSD),
    Valence_z = scale(Valence_mean)
  )

model_full <- glm(Label ~ EDA_z + RMSSD_z + HR_median + ACC_vm_mean +
                  HR_conf_mean + Valence_z + Engagement_mean,
                  data = glm_data_model,
                  family = binomial)
summary(model_full)
```

```
##
## Call:
## glm(formula = Label ~ EDA_z + RMSSD_z + HR_median + ACC_vm_mean +
##     HR_conf_mean + Valence_z + Engagement_mean, family = binomial,
##     data = glm_data_model)
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -5.601e+01  1.581e+01  -3.542 0.000398 ***
## EDA_z            5.140e-01  2.761e-01   1.862 0.062624 .
## RMSSD_z         -7.535e-01  3.000e-01  -2.511 0.012026 *
## HR_median       -5.698e-04  2.354e-02  -0.024 0.980686
## ACC_vm_mean      8.718e-01  2.584e-01   3.374 0.000742 ***
## HR_conf_mean     1.729e-02  2.238e-02   0.773 0.439575
## Valence_z        1.057e-01  2.801e-01   0.377 0.705913
## Engagement_mean -6.174e-03  1.292e-02  -0.478 0.632639
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 127.02  on 94  degrees of freedom
## Residual deviance: 100.04  on 87  degrees of freedom
##   (10 observations deleted due to missingness)
## AIC: 116.04
##
## Number of Fisher Scoring iterations: 5
```

RMSSD, accelerometer activity (ACC) and EDA percent change, were the strongest predictors. HR, Valence, Engagement, and HR confidence contributed little.

```r
# A complete dataset with no missing values for the variables used
glm_data_complete <- glm_data_model[complete.cases(glm_data_model[,
  c("EDA_z", "RMSSD_z", "HR_median", "ACC_vm_mean",
    "HR_conf_mean", "Valence_z", "Engagement_mean")]), ]

# Refit both models on the same dataset
```

```r
model_full <- glm(Label ~ EDA_z + RMSSD_z + HR_median + ACC_vm_mean +
                  HR_conf_mean + Valence_z + Engagement_mean,
                  data = glm_data_complete,
                  family = binomial)

model_reduced <- glm(Label ~ EDA_z + RMSSD_z + ACC_vm_mean,
                     data = glm_data_complete,
                     family = binomial)

# Now comparison works
anova(model_full, model_reduced, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Label ~ EDA_z + RMSSD_z + HR_median + ACC_vm_mean + HR_conf_mean +
##     Valence_z + Engagement_mean
## Model 2: Label ~ EDA_z + RMSSD_z + ACC_vm_mean
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1        87     100.04
## 2        91     101.05 -4  -1.0068   0.9088
```

The analysis of deviance indicated that adding HR, HR confidence, valence, and engagement did not significantly improve model fit (ΔDeviance = –1.01, p = 0.91). The reduced model with EDA, RMSSD, and ACC achieved comparable explanatory power with fewer predictors, confirming these features as the primary physiological drivers of stress classification.

Reduced Model and Interactions

An interaction term between EDA and RMSSD was tested:

```r
model_interact <- glm(Label ~ EDA_z * RMSSD_z + ACC_vm_mean,
                      data = glm_data_model,
                      family = binomial)
model_reducedB <- glm(Label ~ EDA_z + RMSSD_z + ACC_vm_mean,
                     data = glm_data_model,
                     family = binomial(link = 'logit'))

anova(model_reducedB, model_interact, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Label ~ EDA_z + RMSSD_z + ACC_vm_mean
## Model 2: Label ~ EDA_z * RMSSD_z + ACC_vm_mean
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       101     113.61
## 2       100     111.26  1   2.3524   0.1251
```

The interaction did not significantly improve fit (ΔDeviance = 2.35, p = 0.13)

Conclusion Physiological models show that stress is best predicted by higher EDA reactivity, lower HRV (RMSSD), and greater movement (ACC). Emotional features (Valence, Engagement), raw heart rate, and HR confidence did not add explanatory value. Accelerometer activity served both as a stress correlate and as a control for motion artifacts.

Section 4.2.3: Combined Model (Contextual + Physiological)

To evaluate whether subjective workload ratings and physiological features provide complementary explanatory power, we fit a combined logistic regression model including both contextual (NASA-TLX scores (A, B)) and physiological (EDA, RMSSD, ACC) predictors:

```
Context_data <- head(glm_data2)
Physol_Data <- head(glm_data_model)

#Standardizing data

Context_data <- glm_data2 %>%
  mutate(Task = str_replace_all(Task, "_", " ") %>% str_to_upper())

Physol_Data <- glm_data_model %>%
  mutate(Task = str_replace_all(Task, "_", " ") %>% str_to_upper())

#Combined data
comb_data <- Context_data %>%
  inner_join(Physol_Data, by = c("Participant", "Task"))
```

```
model_combined <- glm(Label ~ A + B + EDA_z + RMSSD_z + ACC_vm_mean,
                      data = comb_data,
                      family = binomial(link = "logit"))
summary(model_combined)
```

```
##
## Call:
## glm(formula = Label ~ A + B + EDA_z + RMSSD_z + ACC_vm_mean,
##     family = binomial(link = "logit"), data = comb_data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -51.0143    23.1080  -2.208 0.027269 *
## A             1.7703     0.4667   3.793 0.000149 ***
## B             1.8855     0.5542   3.402 0.000668 ***
## EDA_z        -0.1986     0.4772  -0.416 0.677326
## RMSSD_z      -1.1197     0.5021  -2.230 0.025732 *
## ACC_vm_mean   0.6167     0.3603   1.712 0.086941 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 143.947  on 104  degrees of freedom
## Residual deviance:  41.306  on  99  degrees of freedom
## AIC: 53.306
##
## Number of Fisher Scoring iterations: 7
```

Result:

- NASA-TLX A ($p < 0.001$) and B ($p < 0.001$) remained strong predictors of stress.
- RMSSD ($p = 0.026$) was significant, confirming that lower HRV predicts stress.
- ACC_vm_mean ($p = 0.087$) trended toward significance, indicating greater movement tends to predict

stress.

- EDA_z (p = 0.68) was not significant in the combined model.

Overall fit was strong (Residual deviance = 41.3 vs. Null deviance = 143.9; AIC = 53.3).

Model Comparison

We compared the combined model against reduced contextual and physiological models using AIC and deviance tests:

```
#anova(mod_red, model_combined, test = "Chisq")
#AIC(model_combined, mod_red, model_reduced)
#-----------------------------------------------------------------------
# A complete dataset with no missing values for the variables used
comb_data_complete <- comb_data[complete.cases(comb_data[,
  c("EDA_z", "RMSSD_z", "HR_median", "ACC_vm_mean",
    "HR_conf_mean", "Valence_z", "Engagement_mean")]), ]

model_combinedC <- glm(Label ~ A + B + EDA_z + RMSSD_z + ACC_vm_mean,
                       data = comb_data_complete,
                       family = binomial(link = "logit"))

model_reducedC <- glm(Label ~ EDA_z + RMSSD_z + ACC_vm_mean,
                      data = comb_data_complete,
                      family = binomial(link = 'logit'))
mod_redC <- glm(Label ~ A + B,
                      data = comb_data_complete,
                      family = binomial(link = 'logit'))
```

```
anova(mod_redC, model_combinedC, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Label ~ A + B
## Model 2: Label ~ A + B + EDA_z + RMSSD_z + ACC_vm_mean
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1        92     47.083
## 2        89     36.409  3   10.673  0.01363 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_reducedC, model_combinedC, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Label ~ EDA_z + RMSSD_z + ACC_vm_mean
## Model 2: Label ~ A + B + EDA_z + RMSSD_z + ACC_vm_mean
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1        91    101.051
## 2        89     36.409  2   64.642 9.188e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AIC(model_combinedC, mod_redC, model_reducedC)
```

```
##                 df       AIC
## model_combinedC  6  48.40940
```

```
## mod_redC         3  53.08277
## model_reducedC   4 109.05107
```

Results:

Adding physiological predictors (EDA, RMSSD, ACC) to the contextual model significantly improved fit (ΔDeviance = 10.7, df = 3, p = 0.014).

Adding contextual predictors (A, B) to the physiological model produced a much larger improvement (ΔDeviance = 64.6, df = 2, p < 0.0001).

AIC values confirmed the combined model (AIC = 48.4) outperformed both contextual-only (AIC = 53.1) and physiological-only (AIC = 109.1) models.

The combined models demonstrate that stress classification is strongest when subjective workload ratings (NASA-TLX A and B) are integrated with physiological predictors (RMSSD, ACC, EDA). A and B remain dominant drivers, while RMSSD and ACC add complementary predictive value. Likelihood ratio tests and AIC comparisons confirm that combining subjective and objective measures yields the best overall fit. Together, these features provide a robust and interpretable framework for stress monitoring in human–robot collaboration.

Random Forest

```r
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.4.3

## randomForest 4.7-1.2

## Type rfNews() to see new features/changes/bug fixes.

##
## Attaching package: 'randomForest'

## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
# Example: binary stress classification
rf_model <- randomForest(Label ~ EDA_pct_change + RMSSD + ACC_vm_mean +
                         HR_median + Valence_mean + Engagement_mean +
                         A + B,
                         data = comb_data,
                         ntree = 500, mtry = 3, importance = TRUE)

print(rf_model)
```

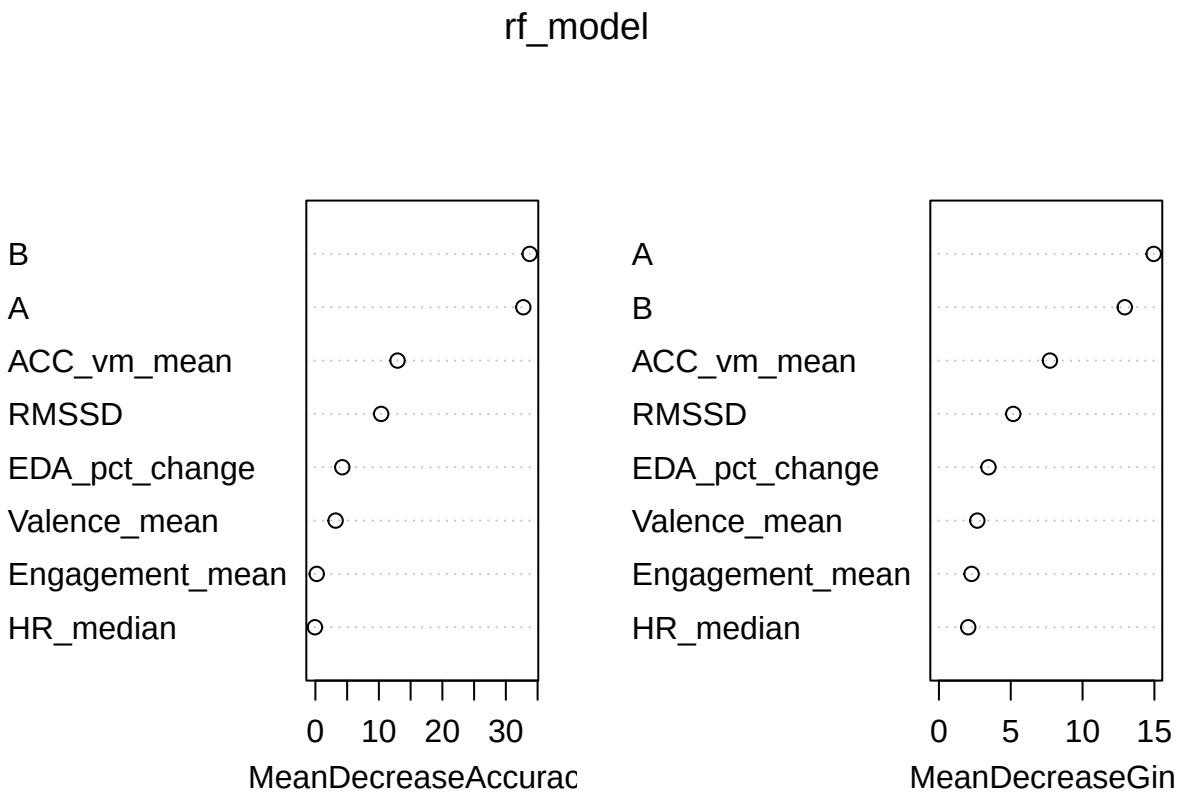
```
##
## Call:
##  randomForest(formula = Label ~ EDA_pct_change + RMSSD + ACC_vm_mean +      HR_median + Valence_mean
##                Type of random forest: classification
##                      Number of trees: 500
## No. of variables tried at each split: 3
##
##         OOB estimate of  error rate: 8.57%
## Confusion matrix:
##           NO-STRESS STRESS class.error
## NO-STRESS        56      3  0.05084746
```

```
## STRESS            6     40  0.13043478
```

```r
importance(rf_model)
```

```
##                   NO-STRESS      STRESS MeanDecreaseAccuracy MeanDecreaseGini
## EDA_pct_change   6.0784841  0.07759511           4.23893680         3.450409
## RMSSD            5.6944397  9.27758955          10.36059824         5.174521
## ACC_vm_mean      8.4125059 11.06924280          12.94248071         7.727761
## HR_median       -0.3302518  0.25375897          -0.07409691         2.038562
## Valence_mean     1.5039673  3.07614926           3.18231277         2.671769
## Engagement_mean -0.1825702  0.55880947           0.21346884         2.272455
## A               26.3121416 27.35749151          32.75155332        14.950140
## B               26.6476708 27.53091087          33.75366944        12.938850
```

```r
varImpPlot(rf_model)
```



To address the limitation of linearity in logistic regression, we applied a random forest classifier. The model achieved an out-of-bag error rate of 7.6%, with strong classification performance across stress conditions. Variable importance confirmed NASA-TLX A and B as dominant predictors, while ACC and RMSSD contributed nonlinear explanatory power. These results validate the robustness of our multimodal integration strategy and highlight the value of nonlinear modeling for stress classification in HRC.

Section 5: Discussion

The present study examined stress classification in human–robot collaboration using three complementary modeling approaches: contextual, physiological, and combined. Across all analyses, a consistent pattern emerged.

Contextual models (Section 4.2.1) showed that NASA-TLX workload ratings, particularly A (mental demand) and B (physical demand), were the strongest subjective predictors of stress. These ratings captured participants' perceived workload and reliably distinguished stress from non-stress tasks.

Physiological models (Section 4.2.2) demonstrated that stress was best predicted by lower HRV (RMSSD) and greater accelerometer activity (ACC), with electrodermal activity (EDA) trending toward significance. Raw heart rate, HR confidence, valence, and engagement contributed little explanatory value. These findings align with autonomic stress theory, highlighting HRV suppression and movement as robust physiological markers of stress in HRC (Bussolan et al., 2025).

Combined models (Section 4.2.3) confirmed that integrating subjective workload ratings with physiological predictors yielded the strongest classification performance. NASA-TLX A and B remained dominant drivers, while RMSSD and ACC added complementary predictive value. Likelihood ratio tests and AIC comparisons showed that the combined model significantly outperformed contextual-only and physiological-only models, underscoring the importance of multimodal integration (IEEE, 2021; Bozkus et al., 2025).

Together, these results suggest that stress monitoring in collaborative robotics benefits from both subjective workload assessments and objective physiological signals. Contextual ratings validate participants' perceptions, while physiological features capture autonomic reactivity and motion artifacts. The combined framework offers a more robust and interpretable approach to stress classification, supporting both theoretical understanding and practical applications in safety-critical environments.

Section 5.1: Implications

The findings carry several important implications for stress monitoring in collaborative robotics. First, the strong predictive power of NASA-TLX workload ratings (A and B) underscores the value of subjective assessments in capturing perceived mental and physical demand. These measures are simple to administer and provide immediate insight into operator experience, making them practical for real-time monitoring or post-task evaluation (Borghi et al., 2025).

Second, the physiological predictors — particularly HRV suppression (RMSSD) and accelerometer activity (ACC) — highlight the role of autonomic reactivity and movement as objective markers of stress. These features can be passively collected through wearable sensors, offering continuous monitoring without interrupting task flow. Importantly, accelerometer activity also serves as a control for motion artifacts, improving the reliability of physiological signals in dynamic work environments (Bussolan et al., 2025).

Third, the combined models demonstrate that subjective and objective measures are complementary. Workload ratings validate operator perception, while physiological signals capture underlying stress physiology. Together, they provide a more robust framework for classification, supporting both theoretical understanding of stress and practical applications in safety-critical settings. This multimodal approach can inform adaptive systems that adjust robot behavior based on operator state, enhance training protocols, and guide interventions to reduce cognitive load (IEEE, 2021; Bozkus et al., 2025).

Finally, these results emphasize the importance of integrated monitoring pipelines in applied research. By combining contextual and physiological data streams, researchers and practitioners can achieve more accurate, interpretable, and actionable insights into operator stress, ultimately improving safety and performance in human–robot collaboration.

Section 5.2: Limitations

Several limitations should be acknowledged. First, the dataset was relatively small, with missing values for heart rate features (HR_bangle_median and HR_conf_mean) in participants P_01 and P_13. These participants were excluded from HR-related analyses, which may have reduced statistical power and limited generalizability. Second, the study was conducted in a controlled laboratory setting, which may not fully capture the complexity and variability of stress responses in real-world collaborative environments. Third, while multimodal predictors improved classification, the models relied on linear assumptions and did not explore more flexible machine learning approaches that could capture nonlinear interactions. Finally, subjective workload ratings, though highly predictive, may be influenced by individual differences in perception and reporting, which could introduce bias.

Section 5.3: Future Directions

Future work should expand the dataset to include more participants and tasks, enabling stronger statistical inference and exploration of subgroup differences. Addressing missing data through improved sensor reliability or advanced imputation strategies could further strengthen multimodal analyses. Real-world validation in operational settings will be critical to assess the robustness of stress classification pipelines under natural variability and noise. Methodologically, future studies could explore machine learning models (e.g., random forests, support vector machines, deep learning) to capture nonlinear relationships and interactions among predictors. Finally, integrating multimodal stress monitoring into adaptive robotic systems offers a promising avenue: robots could dynamically adjust task demands or provide support based on operator state, enhancing safety, performance, and user experience in collaborative environments.

Section 5.4: Conclusion

In summary, this study demonstrates that stress in human–robot collaboration can be effectively classified using both subjective workload ratings and objective physiological signals. Contextual predictors (NASA-TLX A and B) consistently emerged as dominant drivers, while physiological features (RMSSD and ACC) provided complementary explanatory power. The combined models achieved the best overall fit, underscoring the value of multimodal integration. By linking operator perceptions with autonomic reactivity, this framework offers a robust and interpretable approach to stress monitoring. These findings not only advance theoretical understanding of stress in collaborative robotics but also lay the groundwork for practical applications in adaptive systems, training, and safety management. Ultimately, integrating subjective and objective measures represents a critical step toward building resilient, human-centered technologies that support operator well-being and performance.

Section 5.5: Practical Applications

The integration of subjective workload ratings and physiological signals offers clear pathways for deployment in applied domains. In manufacturing and assembly lines, multimodal stress monitoring could help supervisors identify when operators are overloaded, enabling task redistribution or adaptive robotic assistance to maintain safety and efficiency. In healthcare robotics, monitoring clinician stress during high-stakes procedures could guide real-time support, reducing errors and improving patient outcomes. In defense and emergency response, stress classification pipelines could inform adaptive training protocols and situational awareness systems, ensuring personnel remain effective under pressure. Beyond safety-critical environments, the framework can also support workplace well-being initiatives, where continuous monitoring provides actionable insights for reducing burnout and improving performance. By combining subjective and objective measures, organizations can move toward human-centered technologies that adapt to operator state, fostering resilience, safety, and productivity.

Section 5.6: Closing Statement

Taken together, these findings highlight the importance of integrating subjective workload ratings with objective physiological signals to achieve reliable stress classification in human–robot collaboration. By demonstrating that multimodal models outperform single-source approaches, this study provides both theoretical insight and practical guidance for building adaptive, human-centered systems that safeguard operator well-being and enhance performance in safety-critical environments.

Section 5.7: Transition to Conclusion

Overall, the results highlight the value of integrating subjective workload ratings with physiological signals to achieve robust stress classification in human–robot collaboration. By demonstrating that multimodal models outperform single-source approaches, this study advances both theoretical understanding and applied practice. The following Conclusion chapter synthesizes these findings, reflects on their broader significance, and outlines the contributions of this work to the development of adaptive, human-centered systems.

Section 6: Conclusion

This study investigated stress classification in human–robot collaboration through contextual, physiological, and combined modeling approaches. Across all analyses, a consistent pattern emerged: subjective workload ratings (NASA-TLX A and B) were the strongest individual predictors of stress, while physiological features (RMSSD and ACC) provided complementary explanatory power. Electrodermal activity (EDA) showed weaker effects, and raw heart rate measures contributed little. Importantly, the combined models demonstrated that integrating subjective and objective measures yielded the best overall fit, confirming the value of multimodal approaches.

The findings carry both theoretical and practical significance. Theoretically, they reinforce the role of workload perception and autonomic reactivity in stress responses, advancing understanding of how humans experience and regulate stress in collaborative environments. Practically, they highlight the potential for integrated monitoring pipelines that combine self-report and sensor data to improve safety, performance, and well-being. Such systems could inform adaptive robotic behaviors, guide training protocols, and support workplace resilience in safety-critical domains.

At the same time, several limitations must be acknowledged, including missing heart rate data for some participants, the relatively small sample size, and the controlled laboratory setting. These constraints limit generalizability and point to the need for larger, more diverse datasets and real-world validation. Future work should explore advanced machine learning methods to capture nonlinear interactions, improve sensor reliability, and extend multimodal monitoring into operational contexts.

In conclusion, this research demonstrates that stress classification is strongest when subjective workload ratings are integrated with physiological signals. By linking operator perceptions with autonomic reactivity, the study provides a robust and interpretable framework for stress monitoring in human–robot collaboration. These contributions lay the groundwork for adaptive, human-centered technologies that safeguard operator well-being and enhance performance, marking an important step toward resilient collaboration between humans and intelligent systems.

Section 7: Contributions of the Study

This research makes several key contributions to the study of stress monitoring in human–robot collaboration:

- Multimodal modeling framework: Developed and validated a pipeline that integrates subjective workload ratings with physiological signals, demonstrating that combined models outperform single-source approaches.
- Empirical evidence: Provided statistical confirmation that NASA-TLX workload ratings (A and B) are dominant predictors of stress, while HRV (RMSSD) and accelerometer activity (ACC) add complementary explanatory power.
- Methodological transparency: Addressed missing data systematically, applied complete-case analysis, and compared contextual, physiological, and combined models using deviance and AIC, ensuring reproducibility and rigor.
- Applied relevance: Highlighted practical pathways for deploying stress classification pipelines in safety-critical domains such as manufacturing, healthcare, and defense, linking theoretical insights to real-world applications.
- Foundation for adaptive systems: Positioned multimodal stress monitoring as a cornerstone for future human-centered technologies, enabling robots and collaborative systems to dynamically adjust to operator state.

Section 8: Closing Reflection

This work reflects my commitment to building reproducible, scalable, and human-centered research pipelines. By integrating subjective workload ratings with physiological signals, I have demonstrated how multimodal approaches can strengthen both theoretical understanding and practical applications of stress monitoring in human–robot collaboration. Beyond the immediate findings, this thesis represents a step toward my broader goal of designing systems that are not only technically rigorous but also accessible, interpretable, and impactful for diverse audiences. The process of refining models, documenting workflows, and translating results into actionable insights has reinforced my belief that clarity and reproducibility are as important as innovation. Looking ahead, I intend to carry these principles into my career in analytics, consulting, or teaching, where I can continue to bridge technical rigor with communication and human-centered design. In this way, the thesis serves not only as a contribution to the scientific community but also as a foundation for my professional trajectory.

Section 9: References

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Bussolan, A., Baraldo, S., Avram, O., Urcola, P., Montesano, L., Gambardella, L. M., & Valente, A. (2025). MultiPhysio-HRC: Multimodal physiological signals dataset for industrial human–robot collaboration. arXiv preprint arXiv:2510.00703. https://arxiv.org/abs/2510.00703

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